

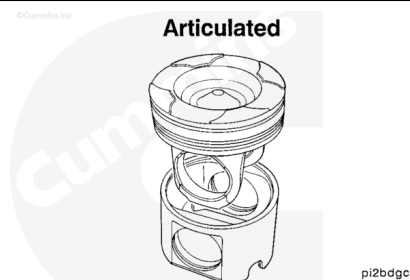
Trainings ▾

General Information

Some Signature™, ISX, and QSX15 engines are equipped with articulated pistons.

The articulated piston is a two-piece piston consisting of a forged steel crown and an aluminum skirt.

An open chamber oil gallery is located on the underside of the crown of the piston to provide more oil contact with the piston and give better piston cooling. Piston cooling nozzle targeting is critical because the oil is captured in shaker trays.



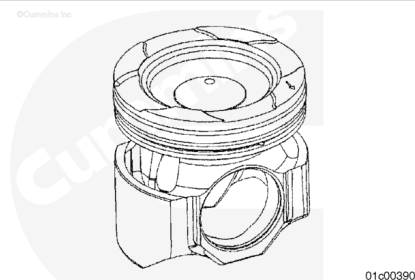
Some engines are equipped with single-piece pistons.

The single-piece piston is a one-piece piston that does **not** have a bushing and is forged from a high-strength, alloy steel.

A cover plate that is **not** removable is located on the underside of the piston to capture oil and cool the piston.

Do **not** remove this cover plate under any circumstances.

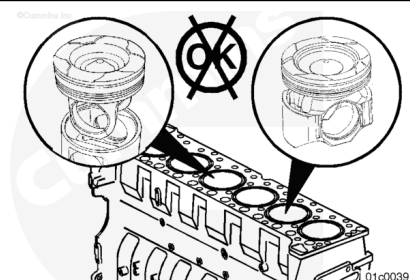
Note : There are two versions of the single-piece piston; the open skirt design and the closed skirt design. These two piston designs can **not** be mixed and matched within an engine.



⚠ CAUTION ⚠

All pistons within a given engine must be of the same design.

All are to be the articulated design, the single piece open skirt design, or the single piece closed skirt design. Do not mix piston designs in the same engine. Single piece pistons must have drilled rods and rod bearings. Articulated pistons can use either drilled or non-drilled rods and bearings.



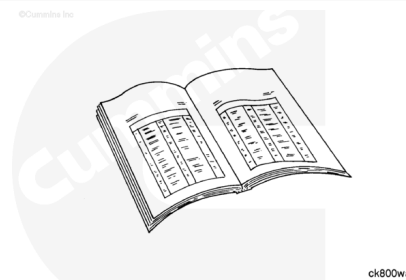
Preparatory Steps

Remove the piston and connecting rod assemblies from the engine. Refer to Procedure 001-054 in Section 1.

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Disassemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.

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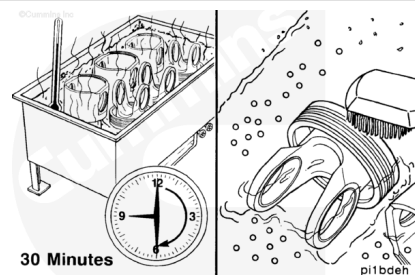


Clean and Inspect for Reuse

Articulated Piston

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



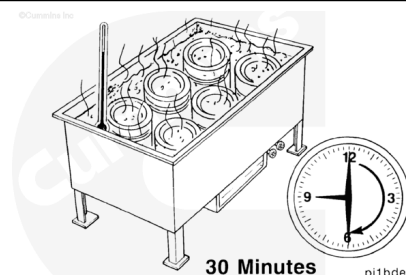
⚠ CAUTION ⚠

To reduce the possibility of damage to the piston crown, the cleaning solvent must be approved for bushing material. If necessary, brush the piston crowns with a soft non-metallic brush. A metallic brush can damage the piston ring grooves.

It is **only** necessary to remove the carbon buildup. It is **not** necessary to make the crown appear to be new.

⚠ CAUTION ⚠

When using walnut shell blast to clean steel piston crowns, use the minimum recommended pressure and do not concentrate the blast in any one area for extended lengths of time. Make sure the pin bore area is protected from the blast so as not to damage the surface finish. Do not use walnut shell blast to clean aluminum piston skirts.



Do **not** use glass or plastic bead media to clean the piston crowns.

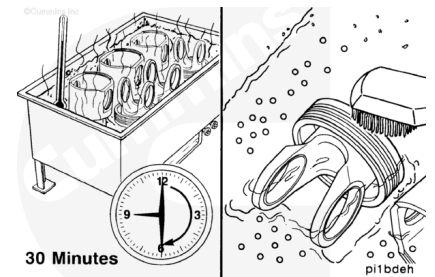
Carbon deposits can be cleaned from the piston crowns by soaking in an approved solution and/or by blasting with walnut shell media.

Soak steel pistons for at least 30 minutes in an approved cleaning solvent for steel and for brushing material. Use a hot, soapy solution with a non-metallic brush to remove carbon deposits.

For best results, soak the pistons several hours or overnight.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



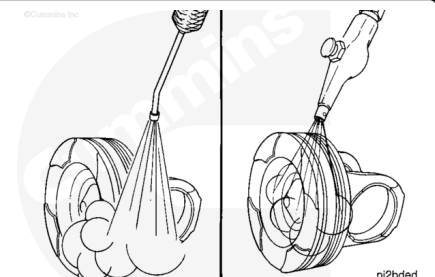
⚠ CAUTION ⚠

If necessary, brush the piston crowns with a soft non-metallic brush. A metallic brush can damage the surface finish. Do not use glass, plastic, or walnut media to clean the aluminum piston skirts. This can damage the pin bore surface finish or prevent the rings from seating correctly in the ring grooves.

Soak aluminum pistons for a minimum of 30 minutes in a tank containing an approved cleaning solvent for aluminum. Use a hot, soapy solution with a non-metallic brush to remove carbon deposits. Anything more abrasive will damage the surface finish.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.

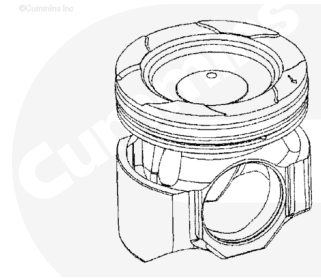
Use steam to clean the pistons.

Dry with compressed air.

Single-Piece Piston

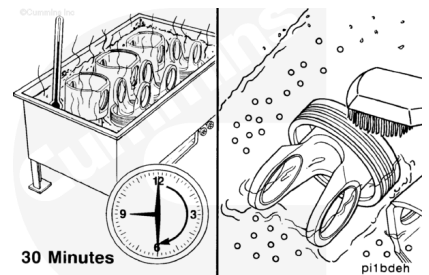
⚠ CAUTION ⚠

Never, under any circumstances, remove the flat, oil cover plate located on the underside of the single-piece pistons.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



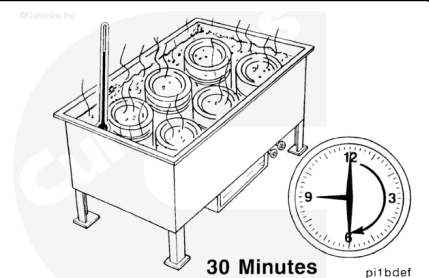
⚠ CAUTION ⚠

To reduce the possibility of damage to the piston, the cleaning solvent must be approved for bushing material. If necessary, brush the pistons with a soft non-metallic brush. A metallic brush can damage the piston ring grooves.

It is **only** necessary to remove the carbon buildup. It is **not** necessary to make the crown appear to be new.

⚠ CAUTION ⚠

When using walnut shell blast to clean steel pistons, use the minimum recommended pressure and do not concentrate the blast in any one area for extended lengths of time. To prevent damage of the surface finish, make sure the pin bore area is protected from the blast.



Do **not** use glass or plastic bead media to clean the piston crowns.

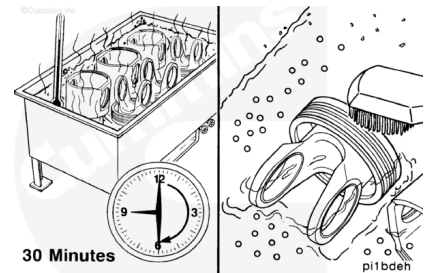
Carbon deposits can be cleaned from the pistons by soaking in an approved solution and/or by blasting with walnut shell media.

Soak steel pistons for at least 30 minutes in an approved cleaning solvent for steel and for brushing material. Use a hot, soapy solution with a non-metallic brush to remove carbon deposits.

For best results, soak the pistons several hours or overnight.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

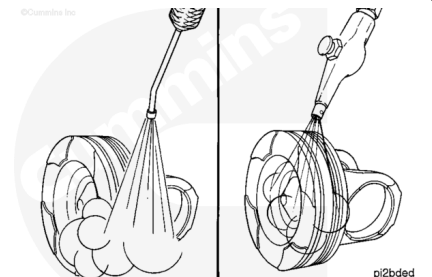


⚠ CAUTION ⚠

If necessary, brush the pistons with a soft non-metallic brush. A metallic brush can damage the surface finish.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.

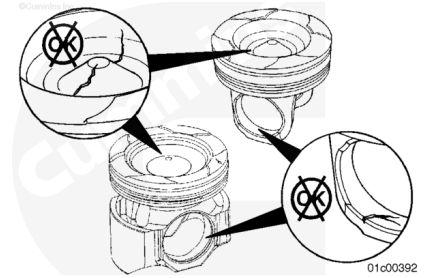
Use steam to clean the pistons.

Dry with compressed air.

Make certain there is **not** walnut shell blast media remaining between the oil cover plate and the piston.

⚠ CAUTION ⚠

Do not use pistons with cracks. If the crown is cracked, it must be replaced. Failure to replace the damaged piston can cause serious engine damage.



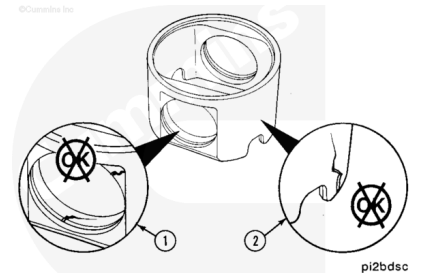
Inspect the piston bowl for cracks or damage.

Inspect the piston pin bore for cracks, scratches, galling, dents, or other damage.

Inspect the piston second land for scuffing. Reference the Single-Piece, Open Skirt Piston Reuse Guidelines, Service Bulletin 4021925.

⚠ CAUTION ⚠

Do not use pistons with cracks. If the skirt is cracked, it must be replaced. Failure to replace the damaged piston can cause serious engine damage.

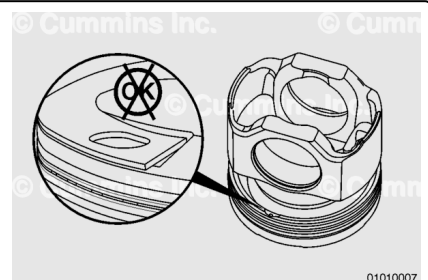


Inspect the piston skirt pin bore (1) and sides (2) for cracks or damage.

Also inspect the skirt for scratches. In general, if the scratches can be felt with a fingernail, do **not** reuse the skirts. Reference the Articulated Piston Skirt Reuse Guidelines for Signature™, ISX, and QSX15 Engines, Service Bulletin 4021468, if additional information is needed.

⚠ CAUTION ⚠

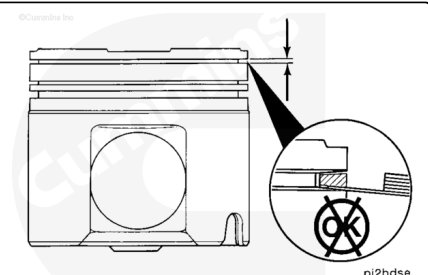
Do not use pistons with loose, broken or missing piston cooling gallery plates. Failure to replace the piston can cause serious engine damage.



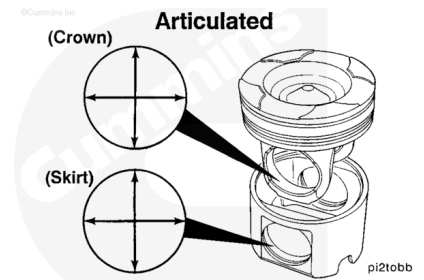
Inspect the piston oil cooling gallery covers for loose, broken, or missing plates. Perform this inspection on new and used pistons.

The ring groove can be inspected with a new ring and a feeler gauge.

Hold a new ring in the groove even with the outside diameter of the piston. Install a 0.15-mm [0.006-in] feeler gauge. If the feeler gauge enters the groove without resistance, there is too much wear. Replace the piston.



Measure the piston pin bore inside diameter on both the crown and skirt on articulated pistons.



Articulated Piston Pin Bore Inside Diameter (Crown)

mm		in
63.56	MIN	2.502
63.58	MAX	2.503

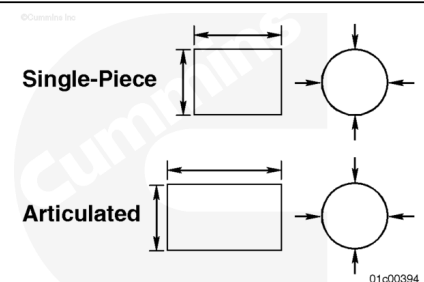
Articulated Piston Pin Bore Inside Diameter (Skirt)

mm		in
63.513	MIN	2.5005
63.525	MAX	2.5010

The piston pin bore inside diameter is **not** a re-use criteria measurement to be taken on single-piece pistons.

Inspect the piston pin for scratches, grooves, or other damage. Do **not** reuse if visually unacceptable.

Measure the piston pin outside diameter. Piston pin outside diameters are the same between articulated and single-piece piston pins.



Piston Pin Outside Diameter

mm		in
63.4670	MIN	2.4987
63.4999	MAX	2.5000

Also measure the piston pin lengths. Single-piece piston pins are shorter than articulated piston pins.

Articulated Piston Pin Length

mm		in
109.5	MIN	4.311
110.0	MAX	4.331

Single-Piece Piston Pin Length

mm		in
84.6	MIN	3.331
85.1	MAX	3.350

Finishing Steps

Note : All engines must be run in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Reference Section 14 Engine Testing for appropriate run-in procedure.

Note : Use the following procedure for general run-in test overview. Refer to Procedure 014-999 in Section F. (</qs3/pubsys2/xml/en/procedures/10/10-014-999.html>)

Note : Use the following procedure if the piston(s) is(are) being replaced due to damaged exhaust valves, intake valves, pistons, or other debris generating failures. Refer to the Prevention of Turbocharger Damage After Engine Mechanical Issue, Service Bulletin 4326040 (</qs3/pubsys2/xml/en/bulletin/4326040.html>)

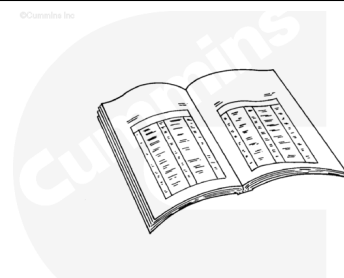
Assemble the piston and connecting rod. Refer to Procedure 001-054 in Section 1.

(</qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html>)

Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.

(</qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html>)

If failure resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)



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